SIH 2026 Prototype — Team Work Assignment

Problem Statement: Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization.

Prototype Scope: A focused 2–3 day MVP demonstrating recovery, sanitization, verification, scoring, comparison, and reporting using controlled test disk images.

1. Team 1 — Data Sanitization & File Sanitization

What the team is: The team responsible for securely removing data and implementing the sanitization layer of the prototype.

What they are building: A sanitization module capable of handling individual files, folders, and controlled disk images, while also classifying the target medium and recommending an appropriate sanitization approach.

Features

- File sanitization/deletion.
- Folder sanitization/deletion.
- Controlled test disk-image sanitization.
- Storage-media classification: HDD, SSD/NVMe, USB/Flash, disk image, and unknown.
- Sanitization-method selection/recommendation based on media type.
- Sanitization status, method, target, and timestamp reporting.
- Safety controls and explicit confirmation before destructive operations.
- SHA-256 hash capture of the target before sanitization.

Physical-device scope: Actual sanitization should be limited to controlled test disk images for the MVP. HDD/SSD/NVMe/USB support should primarily demonstrate classification and method recommendation rather than automatically wiping real devices.

Deliverables: Sanitization engine, media classifier/recommendation module, safety layer, and sanitization result interface.

2. Team 2 — File Recovery

What the team is: The team responsible for finding, reconstructing, recovering, and validating deleted or otherwise recoverable files.

What they are building: A forensic recovery module combining Sleuth Kit filesystem analysis with custom file-carving/recovery logic for the controlled test image.

Features

- Sleuth Kit integration for disk-image and filesystem analysis.
- Filesystem metadata analysis.
- Deleted-file identification.
- Deleted-file recovery using Sleuth Kit tools.
- Custom file carving based on file signatures/headers and appropriate structural markers.

- Recovery of initial target formats such as JPEG, PNG, PDF, DOCX, and ZIP.
- Fragmented-file reconstruction.
- Recovered-file validation and integrity checks.
- Recovery metadata such as file type, location/evidence, size, and recovery status.

Deliverables: Recovery engine, Sleuth Kit wrapper/integration, file-carving module, fragmented-file reconstruction logic, validation module, and recovery results interface.

3. Team 3 — Algorithm / Forensic Intelligence & Verification

What the team is: The team responsible for the algorithms that turn raw recovery and sanitization results into measurable, explainable forensic evidence and verification results.

What they are building: An intelligence and verification layer that evaluates recovery quality, establishes a pre-sanitization baseline, compares results before and after sanitization, verifies sanitization outcomes, maintains a tamper-evident audit trail, and generates the final report/certificate.

Features

- Explainable Recovery Confidence score with reasons supporting the score.
- Project-defined Recoverability Score.
- Pre-sanitization recovery baseline.
- Post-sanitization recovery comparison.
- Sanitization verification.
- Defined verification statement: “No previously identified artifacts were recovered under the defined verification procedure.”
- Before/After PASS or FAIL determination.
- Tamper-evident hash-chained audit trail with integrity verification.
- Automated forensic/sanitization report.
- Sanitization certificate/report output.

AI/ML: AI/ML is intentionally skipped from the working MVP due to the short development window. It can be discussed as a future enhancement during the SIH explanation/presentation.

Deliverables: Confidence and scoring algorithms, baseline/comparison engine, verification module, audit-chain module, and report/certificate generator.

4. Overall Integration

Each team builds its own module and interface, but all three teams follow the same design concepts, visual language, data structures, and integration contracts. The project lead connects the modules into one unified prototype.

Stage	Responsible Team	Main Output
Analyze test image	Team 2	Filesystem/evidence information
Find & recover deleted files	Team 2	Recovered artifacts
Validate recovery	Team 2	Validated recovery results
Calculate confidence/score	Team 3	Confidence + recoverability metrics
Create baseline	Team 3	Pre-sanitization evidence baseline
Sanitize target	Team 1	Sanitization result
Recover again	Team 2	Post-sanitization recovery results
Verify & compare	Team 3	Before/after verification
Generate report	Team 3	Tamper-evident report/certificate

Integration Flow

TEST DISK IMAGE → ANALYZE → FIND DELETED FILES → RECOVER → VALIDATE → CALCULATE RECOVERY CONFIDENCE → CREATE BASELINE → SANITIZE → RECOVER AGAIN → COMPARE BEFORE/AFTER → VERIFY → GENERATE TAMPER-EVIDENT REPORT

Integration principle: Team 2 provides recovery evidence, Team 1 performs sanitization, and Team 3 evaluates and verifies the complete closed-loop process.

5. Project USPs

Primary USP	Don’t just erase data. Prove what was recoverable before sanitization — and prove what remains afterward.
Explainable Recovery Confidence	A recovery score supported by understandable evidence such as valid file signatures, structural checks, parser validation, and filesystem evidence.
Recoverability Score	A project-defined metric that summarizes how recoverable an artifact is rather than simply reporting that a file was found.
Before/After Verification	The system establishes a recovery baseline before sanitization and compares it with post-sanitization recovery results.
Tamper-Evident Audit Trail	Hash-chained audit events make changes to the recorded workflow detectable. This is an audit-integrity mechanism, not blockchain.
Automated Reporting & Certificate	The system produces a consolidated result showing target information, sanitization details, recovery findings, verification outcome, and audit integrity.

Prototype Safety Boundary

The MVP should operate on controlled test disk images by default. It must never automatically target system disks such as /dev/sda or /dev/nvme*. Destructive operations require explicit confirmation, and the target path and SHA-256 should be shown before sanitization. The prototype should not claim universal or permanent deletion, particularly for SSD/NVMe storage.